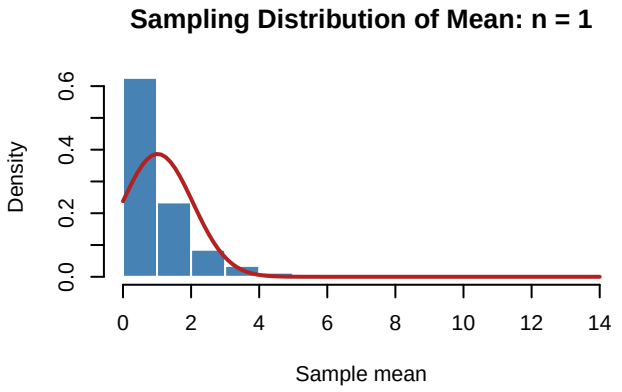
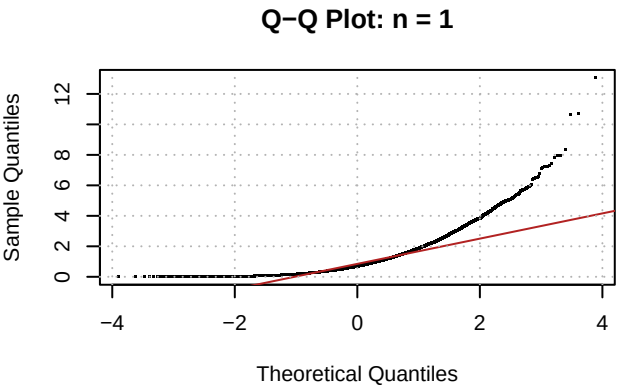
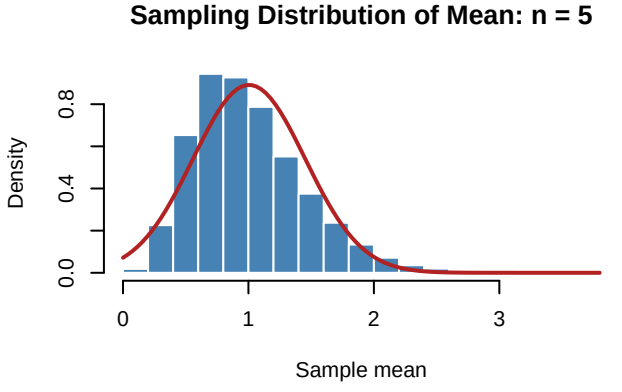
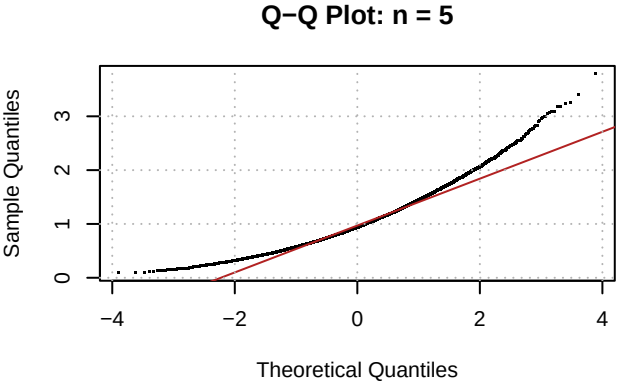
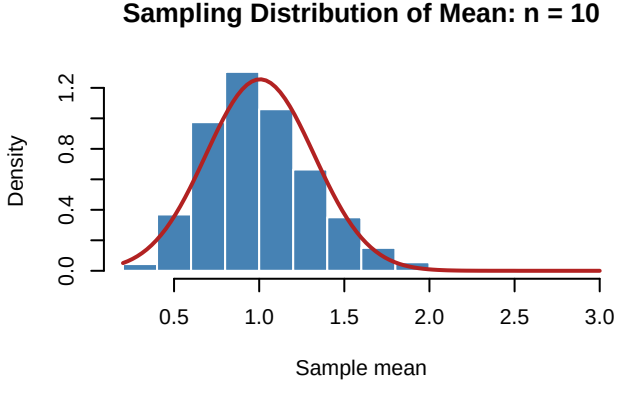
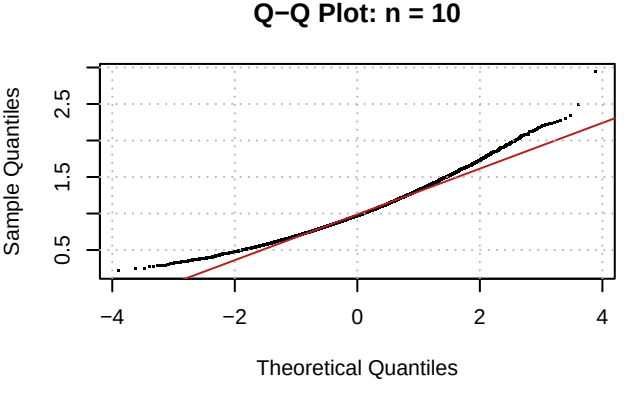

Sampling Distribution of Mean: n = 1
Density
0.0
0.2
0.4
0.6
0
2
4
6
8
10
12
14
Sample mean

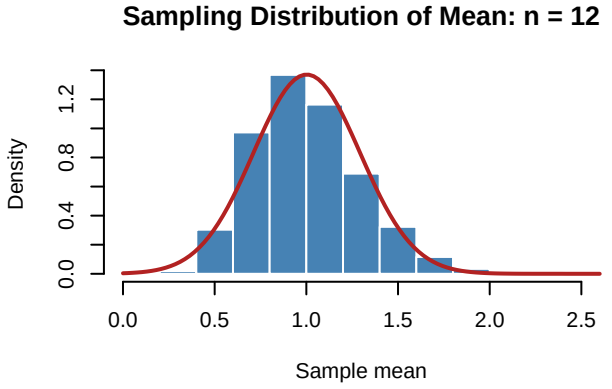

Q−Q Plot: n = 1
Sample Quantiles
0
2
4
6
8
12
−4
−2
0
2
4
Theoretical Quantiles

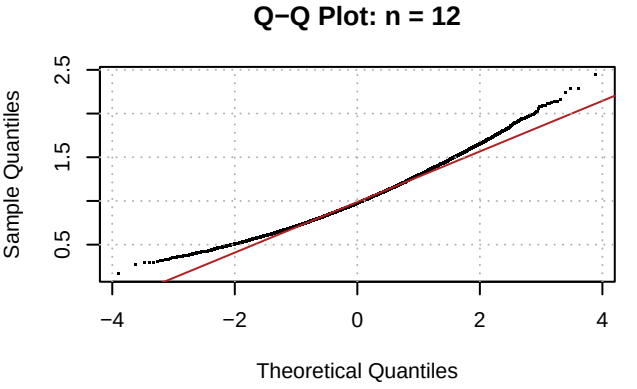

Sampling Distribution of Mean: n = 5
Density
0.0
0.4
0.8
0
1
2
3
Sample mean

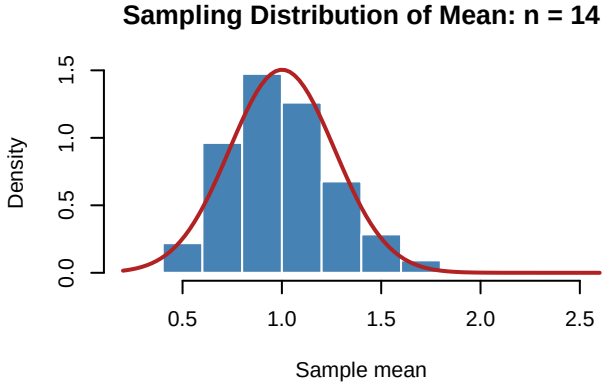

Q−Q Plot: n = 5
Sample Quantiles
0
1
2
3
−4
−2
0
2
4
Theoretical Quantiles

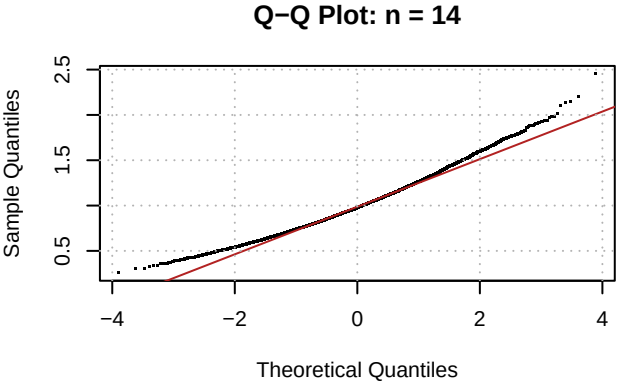

Sampling Distribution of Mean: n = 10
Density
0.0
0.4
0.8
1.2
0.5
1.0
1.5
2.0
2.5
3.0
Sample mean

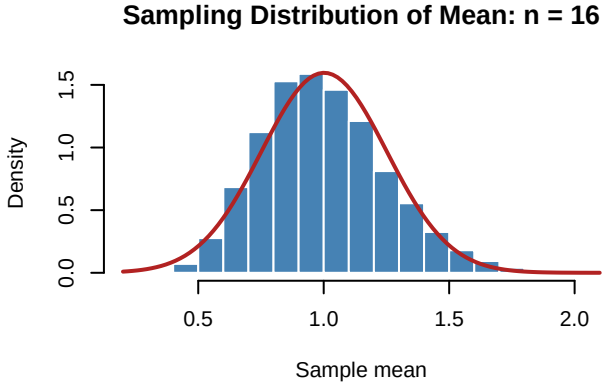

Q−Q Plot: n = 10
Sample Quantiles
0.5
1.5
2.5
−4
−2
0
2
4
Theoretical Quantiles

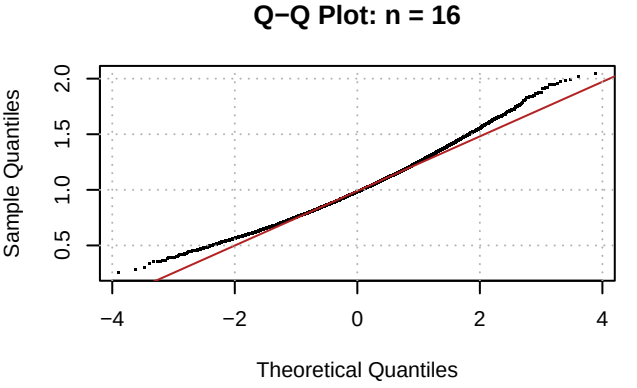

Sampling Distribution of Mean: n = 12
Density
0.0
0.4
0.8
1.2
0.0
0.5
1.0
1.5
2.0
2.5
Sample mean

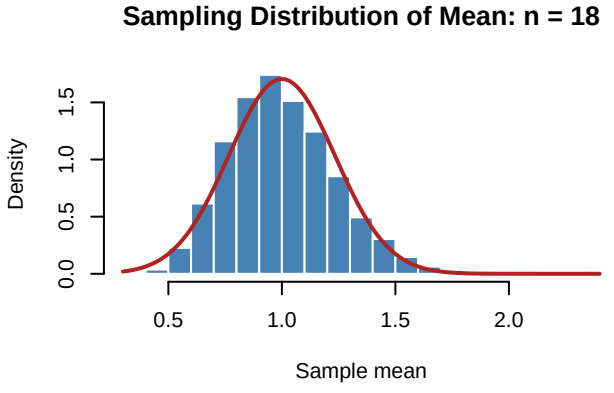

Q−Q Plot: n = 12
Sample Quantiles
0.5
1.5
2.5
−4
−2
0
2
4
Theoretical Quantiles

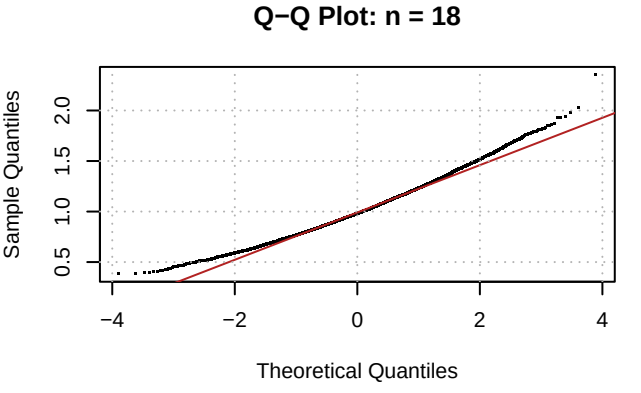

Sampling Distribution of Mean: n = 14
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
2.5
Sample mean

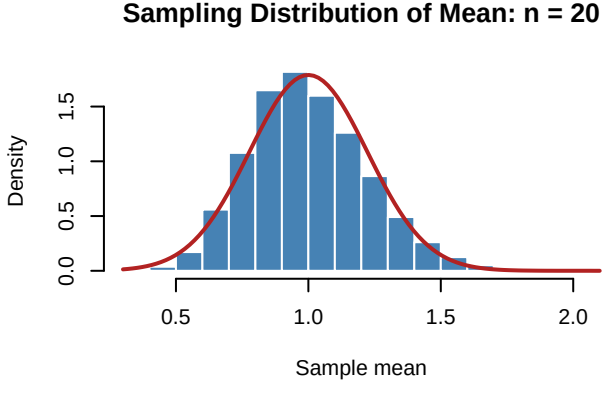

Q−Q Plot: n = 14
Sample Quantiles
0.5
1.5
2.5
−4
−2
0
2
4
Theoretical Quantiles

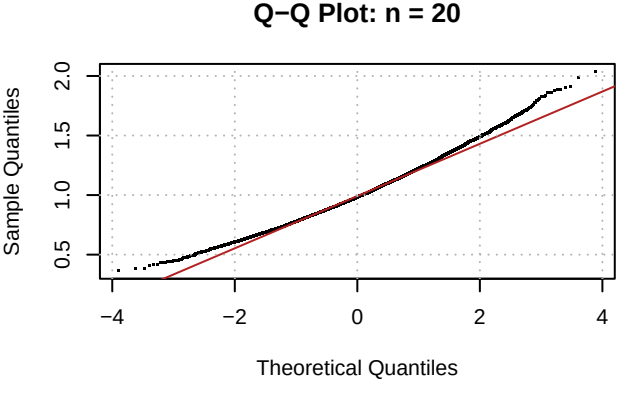

Sampling Distribution of Mean: n = 16
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample mean

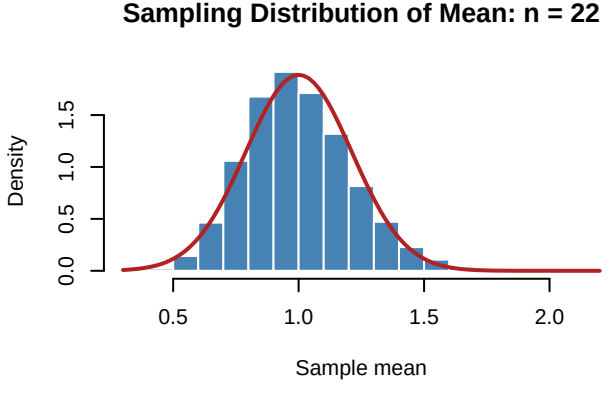

Q−Q Plot: n = 16
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

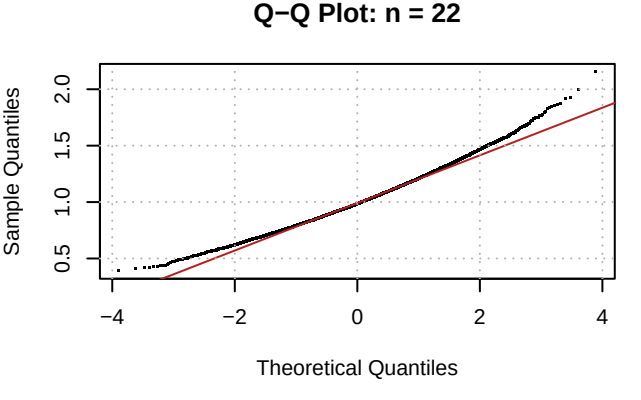

Sampling Distribution of Mean: n = 18
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample mean

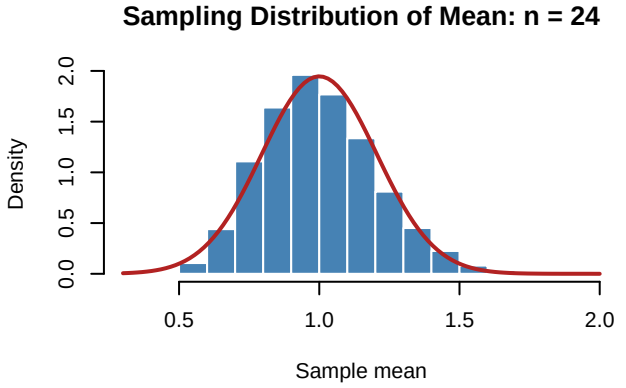

Q−Q Plot: n = 18
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

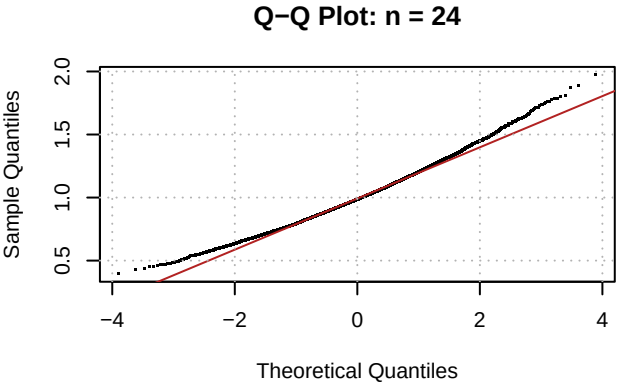

Sampling Distribution of Mean: n = 20
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample mean

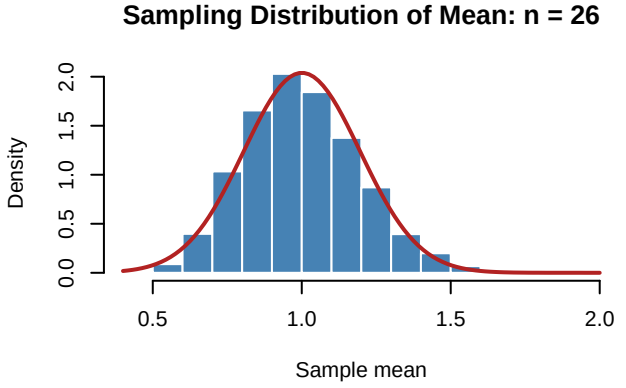

Q−Q Plot: n = 20
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

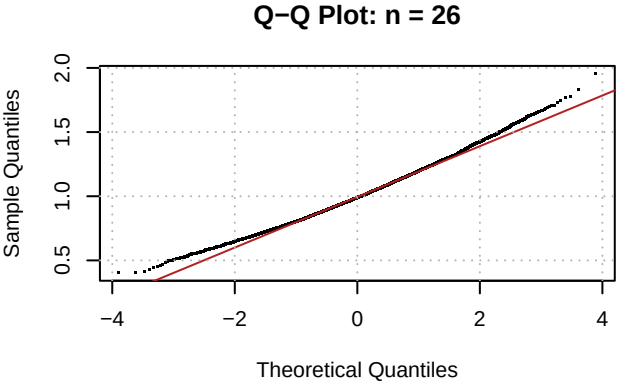

Sampling Distribution of Mean: n = 22
Density
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
Sample mean

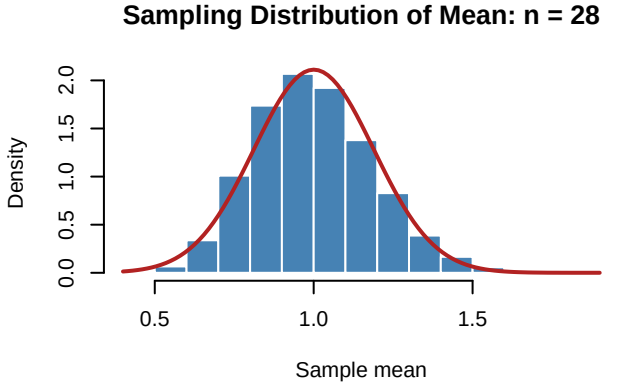

Q−Q Plot: n = 22
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

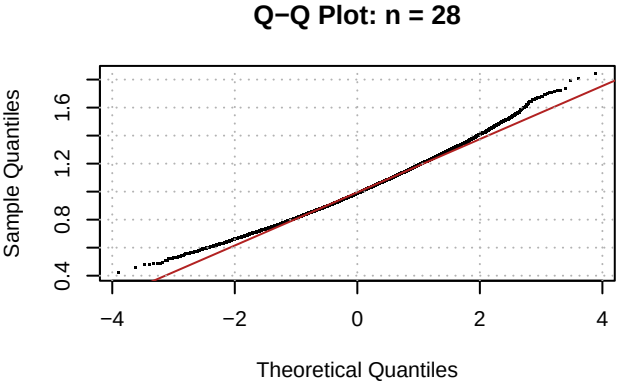

Sampling Distribution of Mean: n = 24
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
2.0
Sample mean

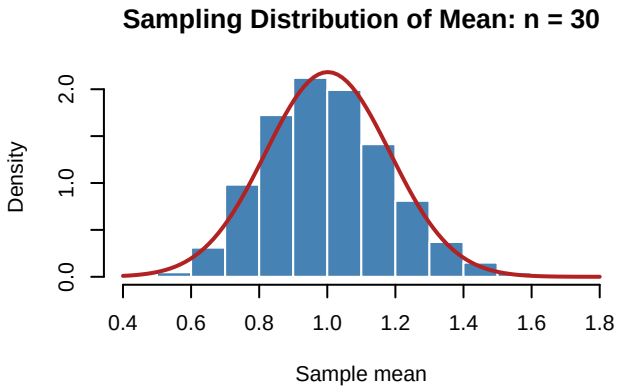

Q−Q Plot: n = 24
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

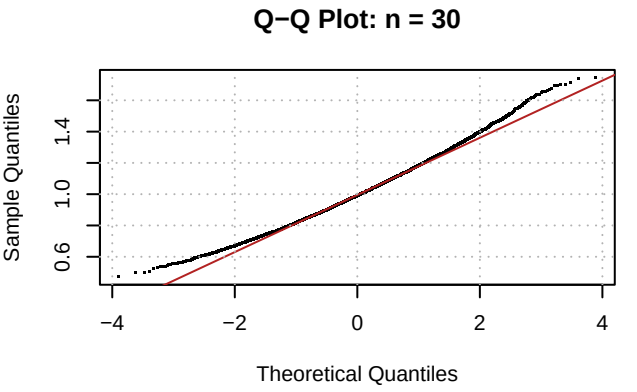

Sampling Distribution of Mean: n = 26
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
2.0
Sample mean

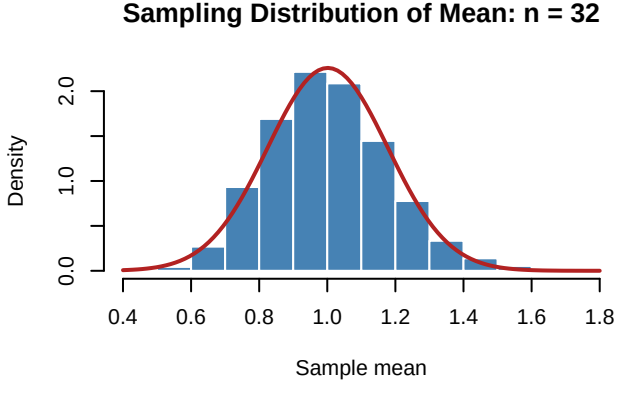

Q−Q Plot: n = 26
Sample Quantiles
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

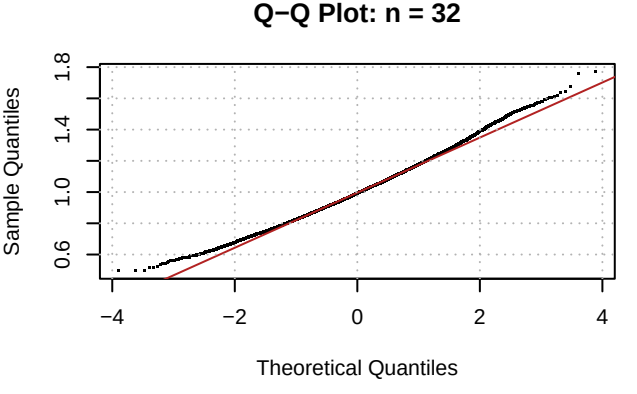

Sampling Distribution of Mean: n = 28
Density
0.0
0.5
1.0
1.5
2.0
0.5
1.0
1.5
Sample mean

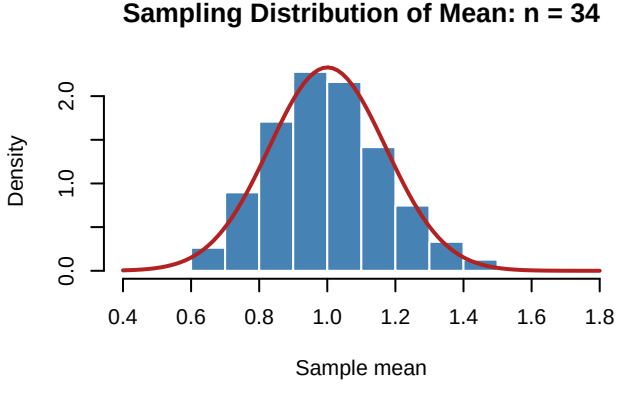

Q−Q Plot: n = 28
Sample Quantiles
0.4
0.8
1.2
1.6
−4
−2
0
2
4
Theoretical Quantiles

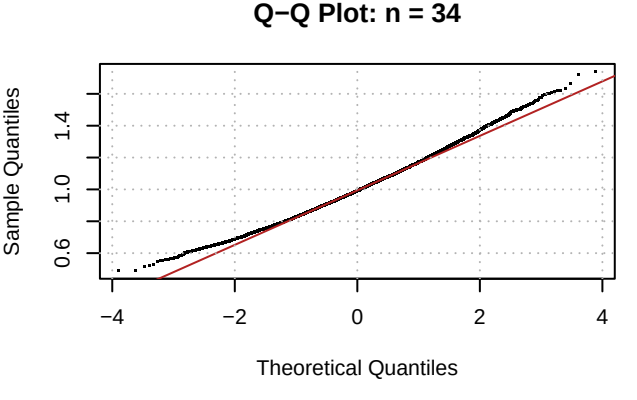

Sampling Distribution of Mean: n = 30
Density
0.0
1.0
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

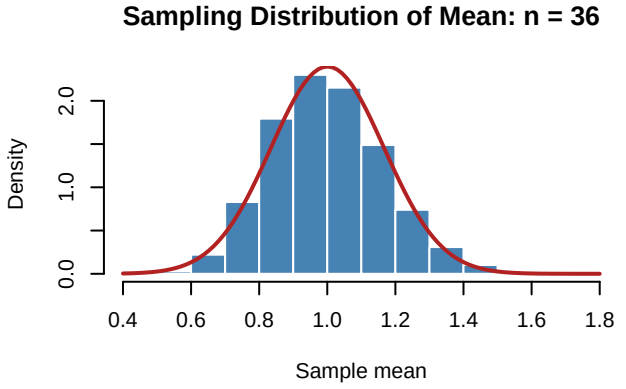

Q−Q Plot: n = 30
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

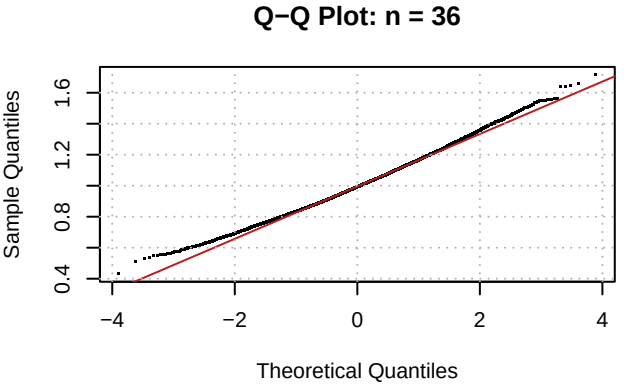

Sampling Distribution of Mean: n = 32
Density
0.0
1.0
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

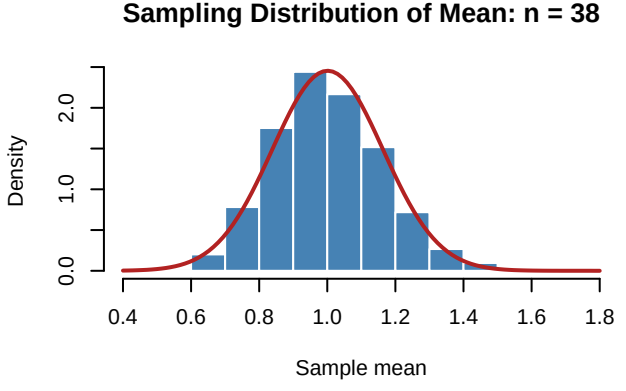

Q−Q Plot: n = 32
Sample Quantiles
0.6
1.0
1.4
1.8
−4
−2
0
2
4
Theoretical Quantiles

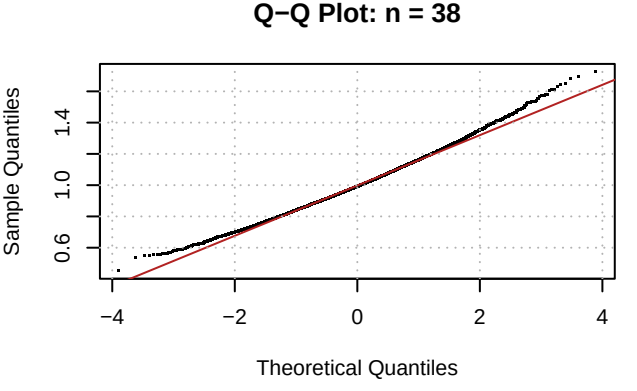

Sampling Distribution of Mean: n = 34
Density
0.0
1.0
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

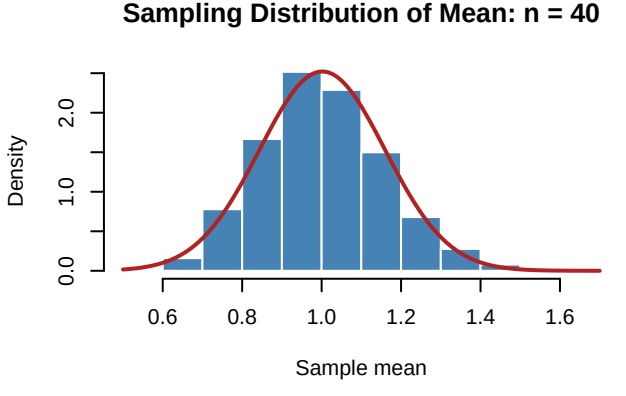

Q−Q Plot: n = 34
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

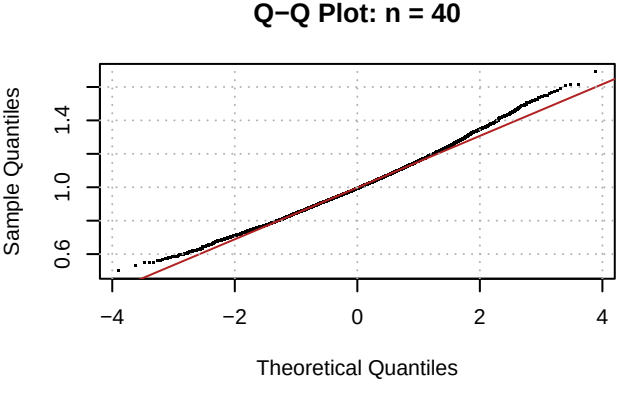

Sampling Distribution of Mean: n = 36
Density
0.0
1.0
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

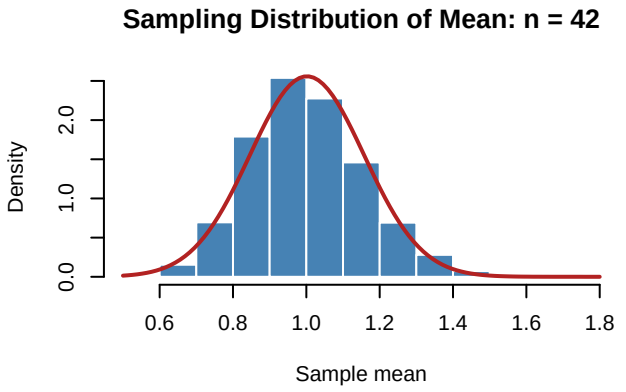

Q−Q Plot: n = 36
Sample Quantiles
0.4
0.8
1.2
1.6
−4
−2
0
2
4
Theoretical Quantiles

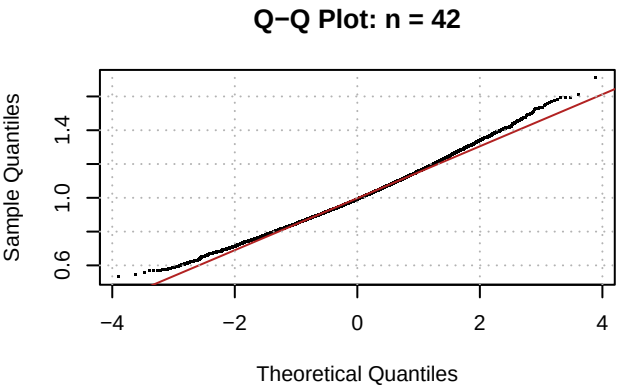

Sampling Distribution of Mean: n = 38
Density
0.0
1.0
2.0
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

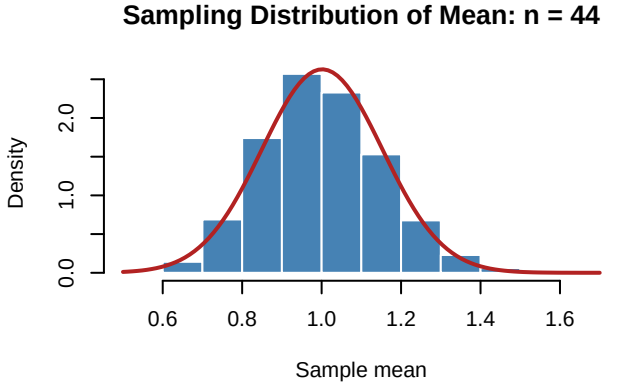

Q−Q Plot: n = 38
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

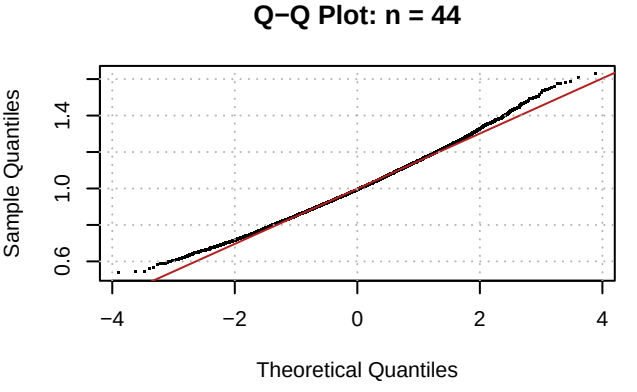

Sampling Distribution of Mean: n = 40
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

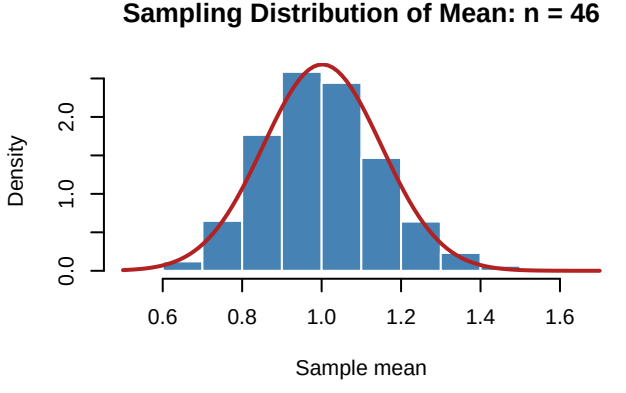

Q−Q Plot: n = 40
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

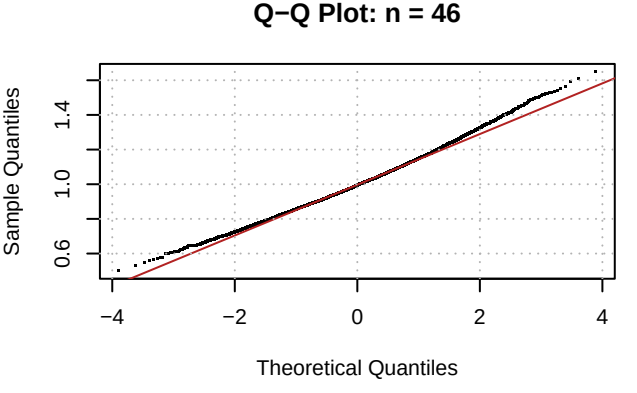

Sampling Distribution of Mean: n = 42
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Sample mean

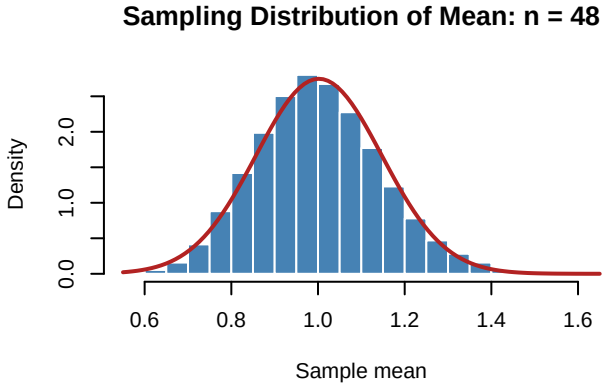

Q−Q Plot: n = 42
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

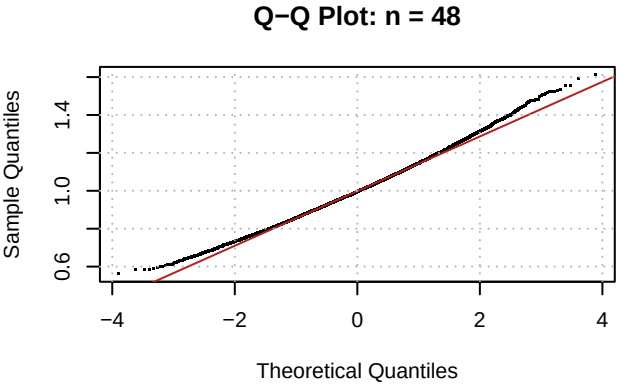

Sampling Distribution of Mean: n = 44
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

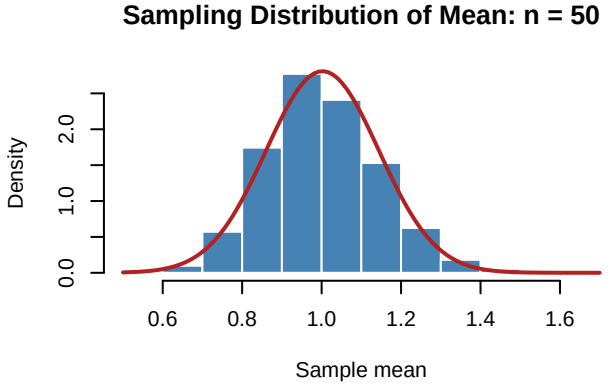

Q−Q Plot: n = 44
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

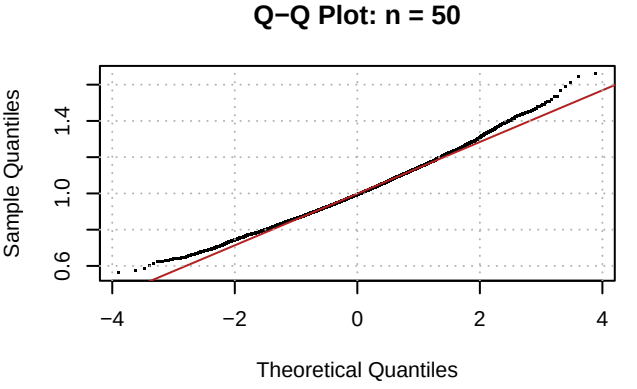

Sampling Distribution of Mean: n = 46
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

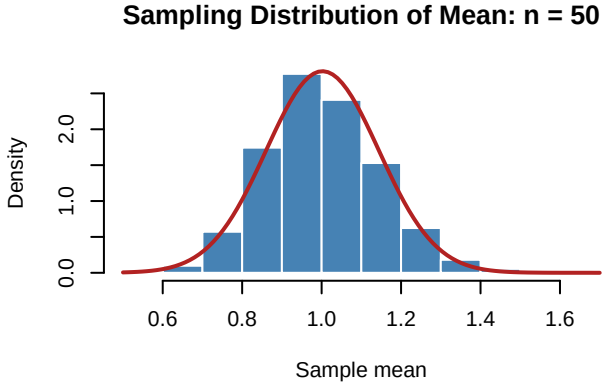

Q−Q Plot: n = 46
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

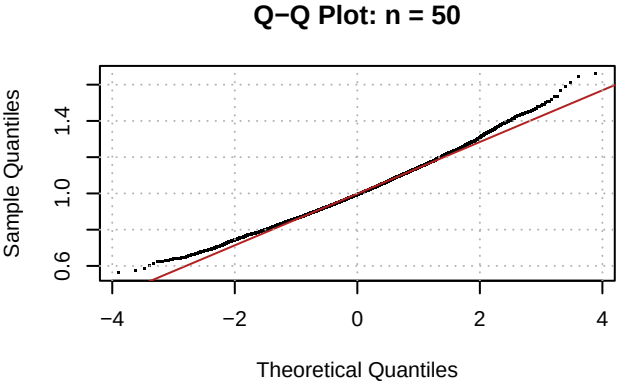

Sampling Distribution of Mean: n = 48
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

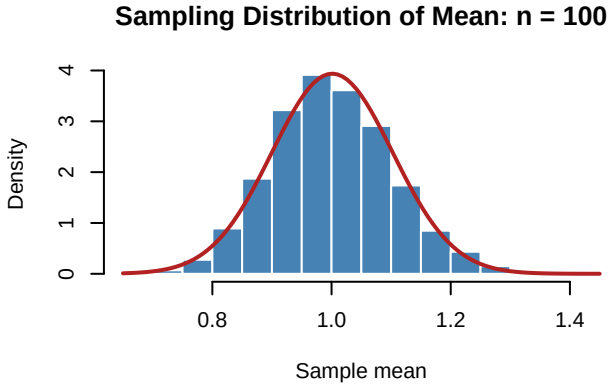

Q−Q Plot: n = 48
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

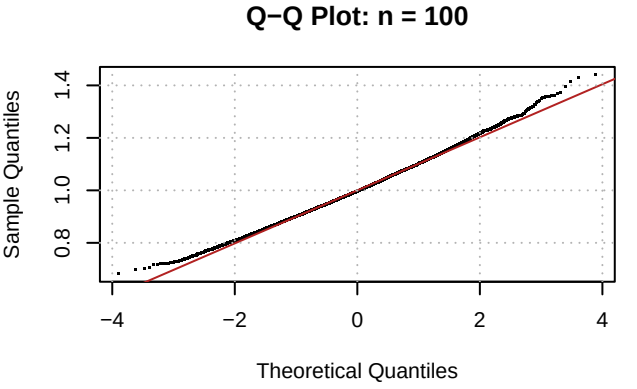

Sampling Distribution of Mean: n = 50
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

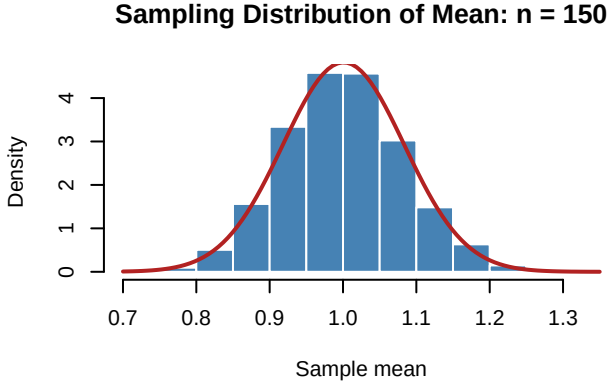

Q−Q Plot: n = 50
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

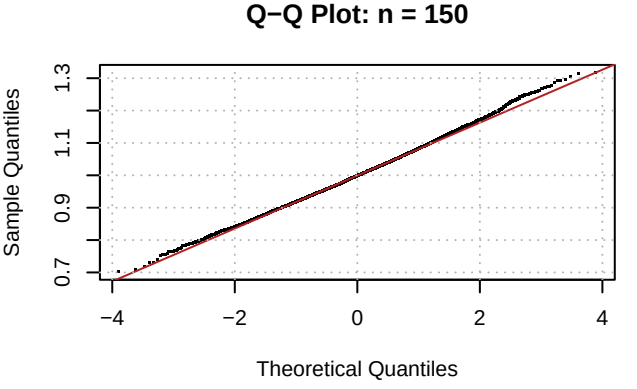

Sampling Distribution of Mean: n = 50
Density
0.0
1.0
2.0
0.6
0.8
1.0
1.2
1.4
1.6
Sample mean

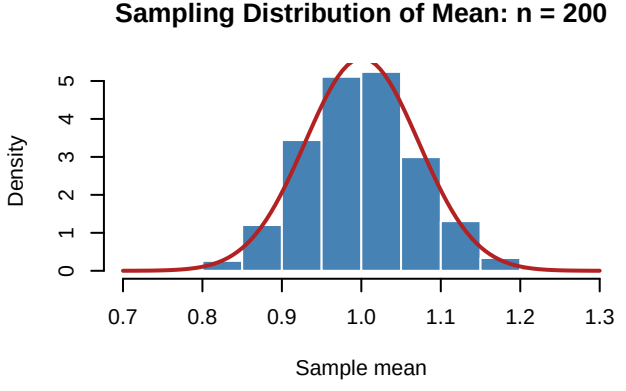

Q−Q Plot: n = 50
Sample Quantiles
0.6
1.0
1.4
−4
−2
0
2
4
Theoretical Quantiles

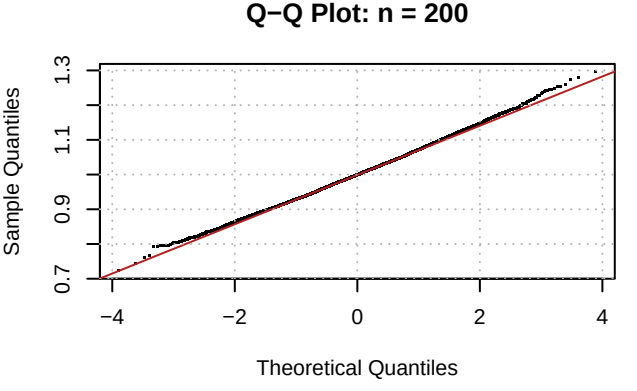

Sampling Distribution of Mean: n = 100
Density
0
1
2
3
4
0.8
1.0
1.2
1.4
Sample mean

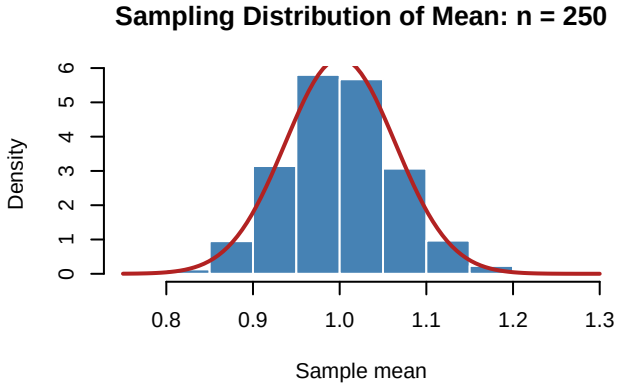

Q−Q Plot: n = 100
Sample Quantiles
0.8
1.0
1.2
1.4
−4
−2
0
2
4
Theoretical Quantiles

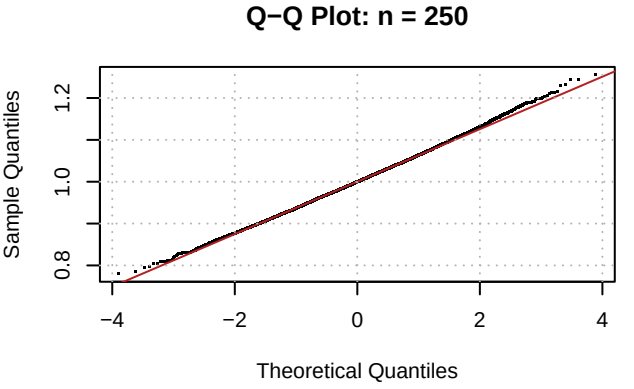

Sampling Distribution of Mean: n = 150
Density
0
1
2
3
4
0.7
0.8
0.9
1.0
1.1
1.2
1.3
Sample mean

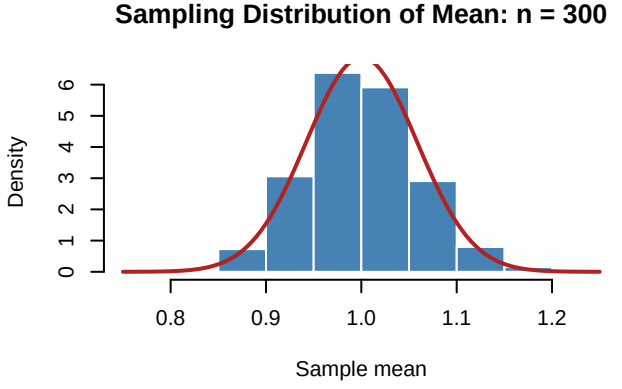

Q−Q Plot: n = 150
Sample Quantiles
0.7
0.9
1.1
1.3
−4
−2
0
2
4
Theoretical Quantiles

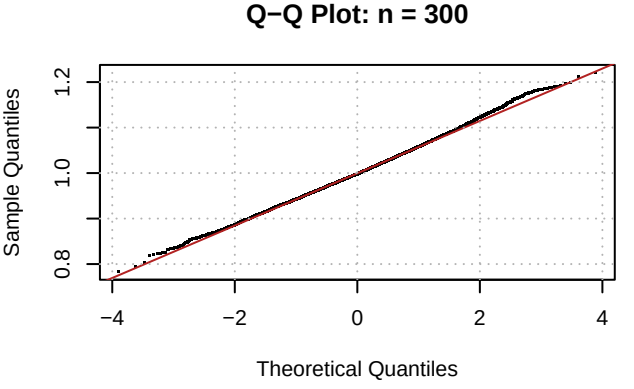

Sampling Distribution of Mean: n = 200
Density
0
1
2
3
4
5
0.7
0.8
0.9
1.0
1.1
1.2
1.3
Sample mean

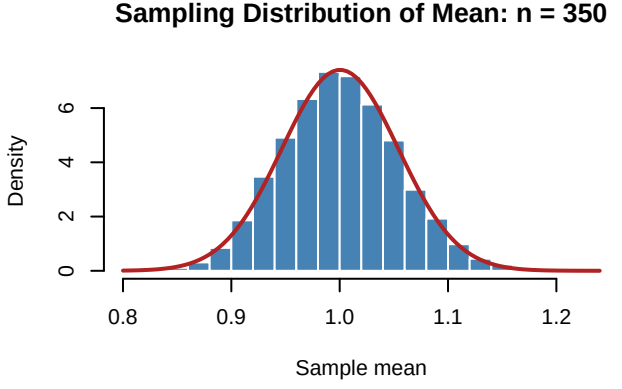

Q−Q Plot: n = 200
Sample Quantiles
0.7
0.9
1.1
1.3
−4
−2
0
2
4
Theoretical Quantiles

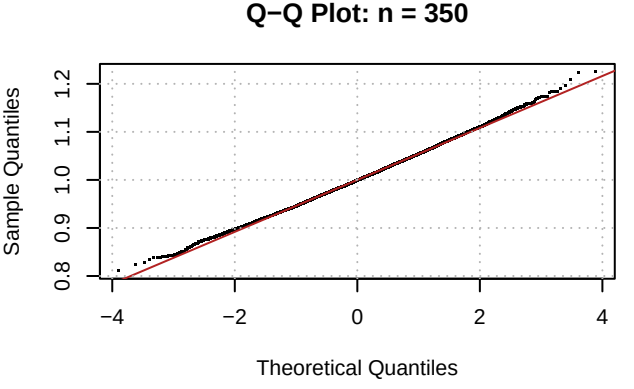

Sampling Distribution of Mean: n = 250
Density
0 1 2 3 4 5 6
0.8 0.9 1.0 1.1 1.2 1.3
Sample mean

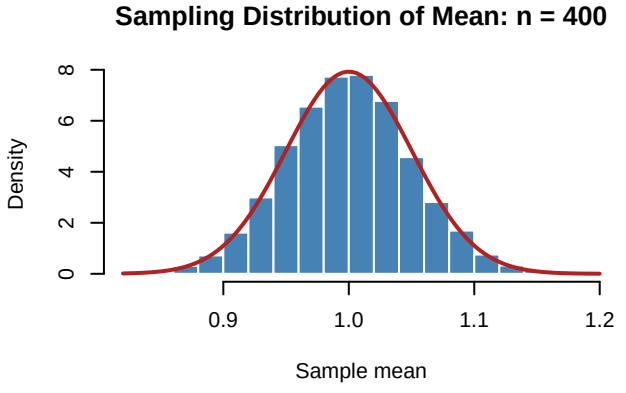

Q−Q Plot: n = 250
Sample Quantiles
0.8 1.0 1.2
−4 −2 0 2 4
Theoretical Quantiles

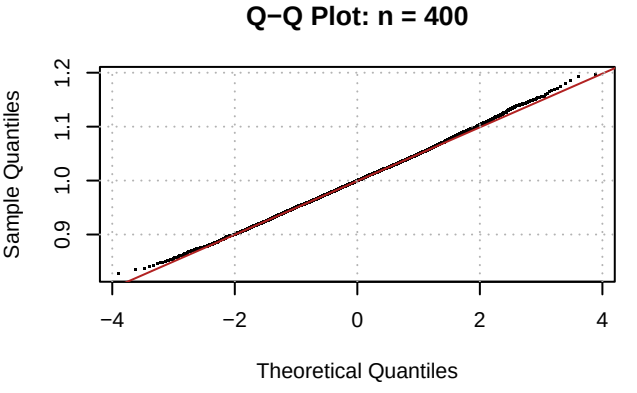

Sampling Distribution of Mean: n = 300
Density
0 1 2 3 4 5 6
0.8 0.9 1.0 1.1 1.2
Sample mean

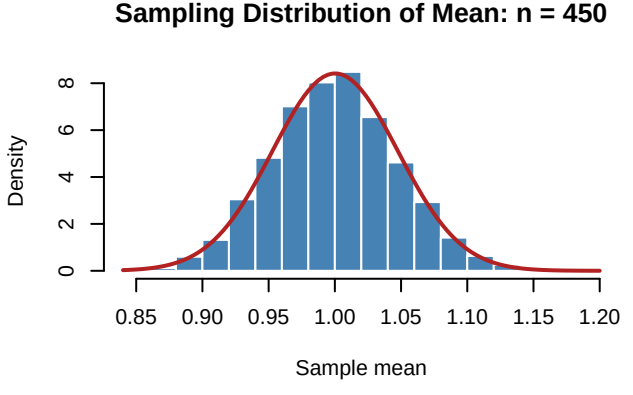

Q−Q Plot: n = 300
Sample Quantiles
0.8 1.0 1.2
−4 −2 0 2 4
Theoretical Quantiles

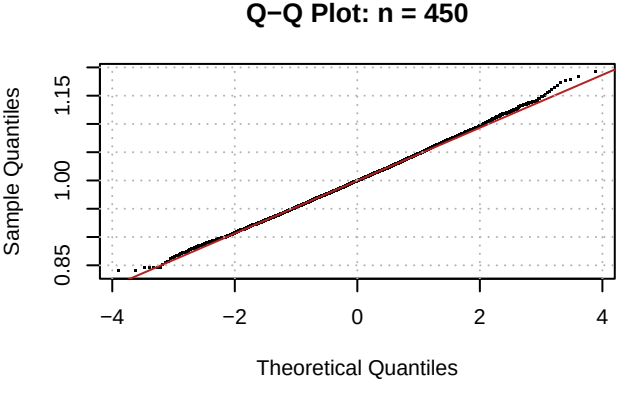

Sampling Distribution of Mean: n = 350
Density
0 2 4 6
0.8 0.9 1.0 1.1 1.2
Sample mean

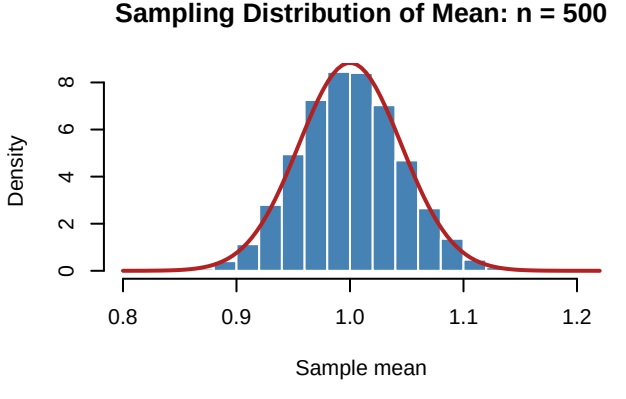

Q−Q Plot: n = 350
Sample Quantiles
0.8 0.9 1.0 1.1 1.2
−4 −2 0 2 4
Theoretical Quantiles

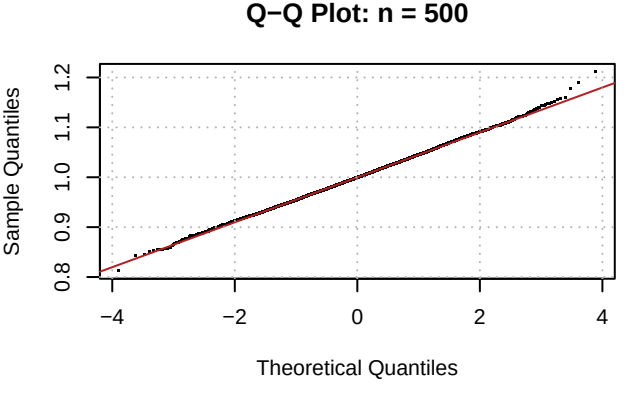

Sampling Distribution of Mean: n = 400
Density
0
2
4
6
8
0.9
1.0
1.1
1.2
Sample mean

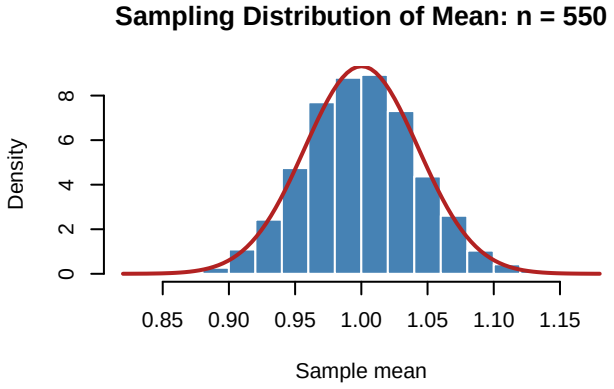

Q−Q Plot: n = 400
Sample Quantiles
0.9
1.0
1.1
1.2
−4
−2
0
2
4
Theoretical Quantiles

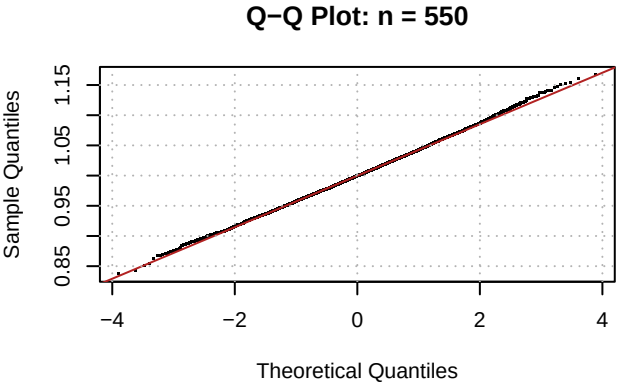

Sampling Distribution of Mean: n = 450
Density
0
2
4
6
8
0.85
0.90
0.95
1.00
1.05
1.10
1.15
1.20
Sample mean

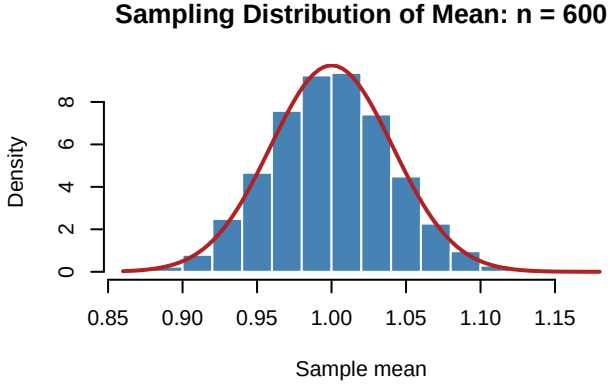

Q−Q Plot: n = 450
Sample Quantiles
0.85
1.00
1.15
−4
−2
0
2
4
Theoretical Quantiles

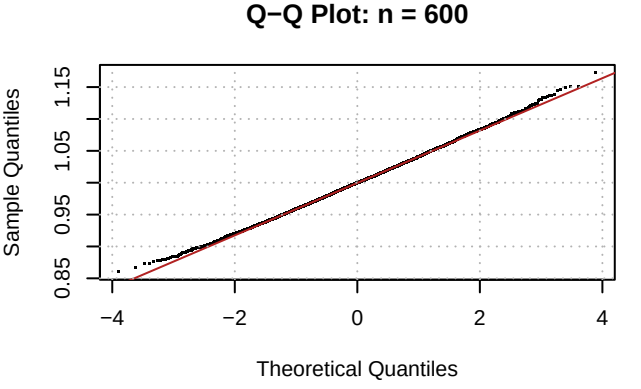

Sampling Distribution of Mean: n = 500
Density
0
2
4
6
8
0.8
0.9
1.0
1.1
1.2
Sample mean

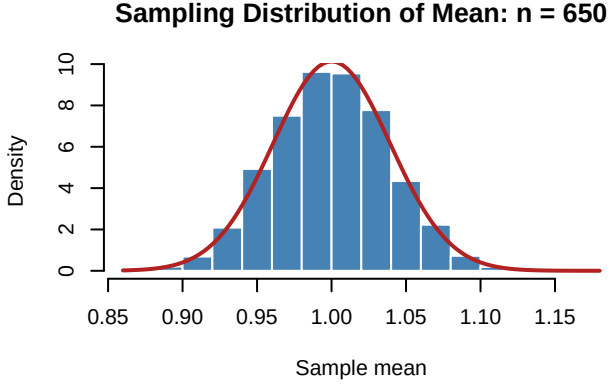

Q−Q Plot: n = 500
Sample Quantiles
0.8
0.9
1.0
1.1
1.2
−4
−2
0
2
4
Theoretical Quantiles

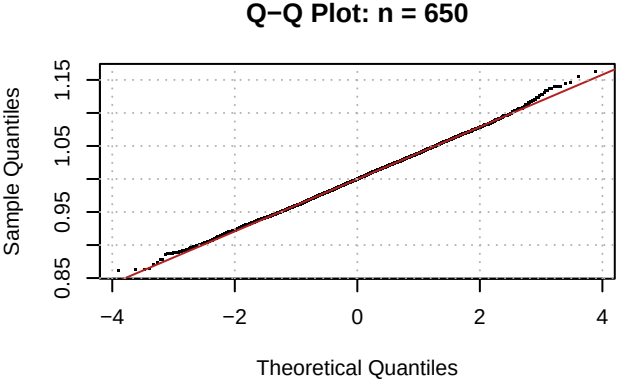

Sampling Distribution of Mean: n = 550
Density
0
2
4
6
8
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

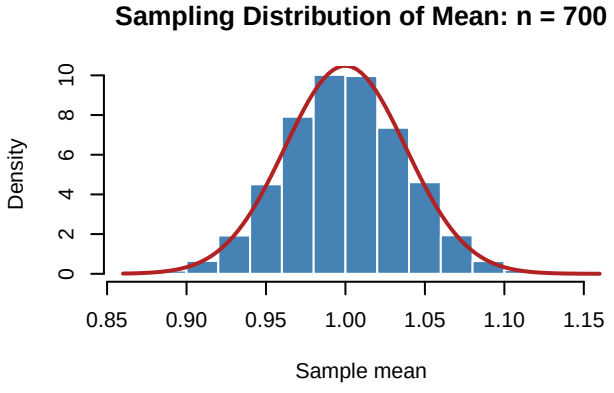

Q−Q Plot: n = 550
Sample Quantiles
0.85
0.95
1.05
1.15
−4
−2
0
2
4
Theoretical Quantiles

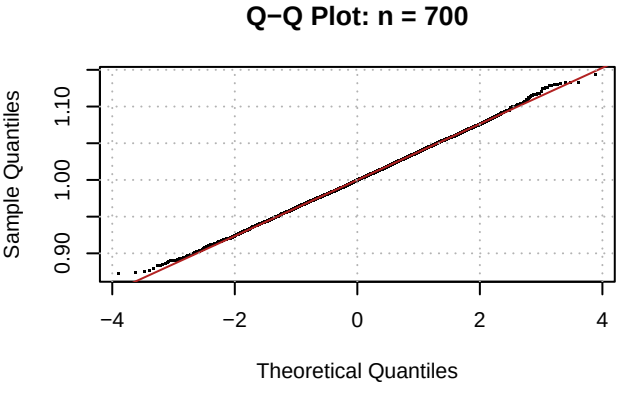

Sampling Distribution of Mean: n = 600
Density
0
2
4
6
8
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

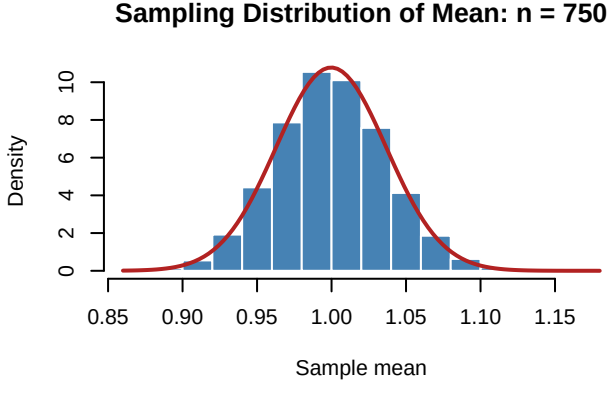

Q−Q Plot: n = 600
Sample Quantiles
0.85
0.95
1.05
1.15
−4
−2
0
2
4
Theoretical Quantiles

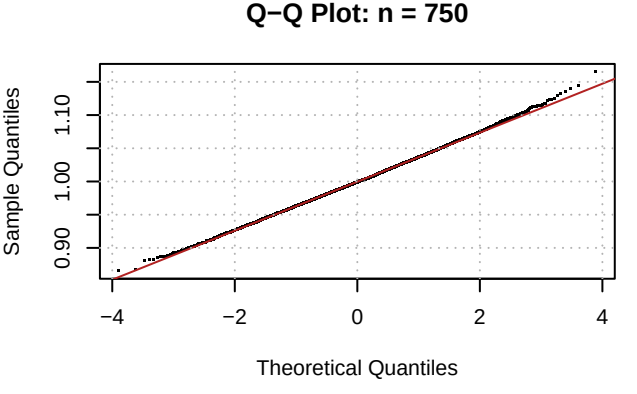

Sampling Distribution of Mean: n = 650
Density
0
2
4
6
8
10
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

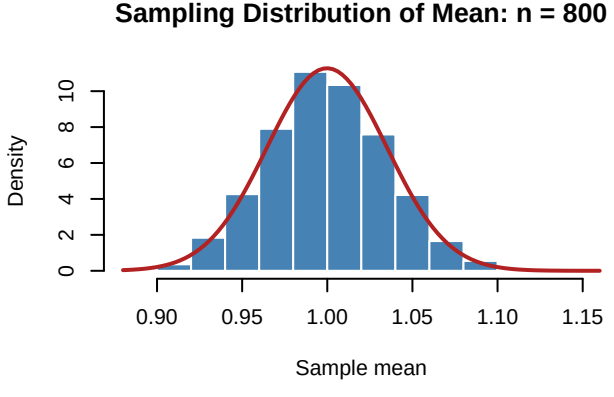

Q−Q Plot: n = 650
Sample Quantiles
0.85
0.95
1.05
1.15
−4
−2
0
2
4
Theoretical Quantiles

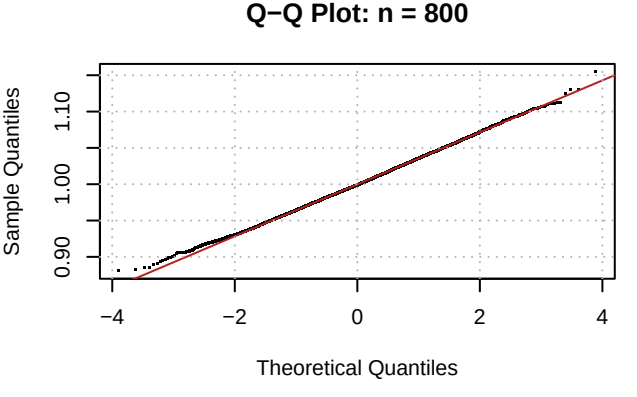

Sampling Distribution of Mean: n = 700
Density
0
2
4
6
8
10
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

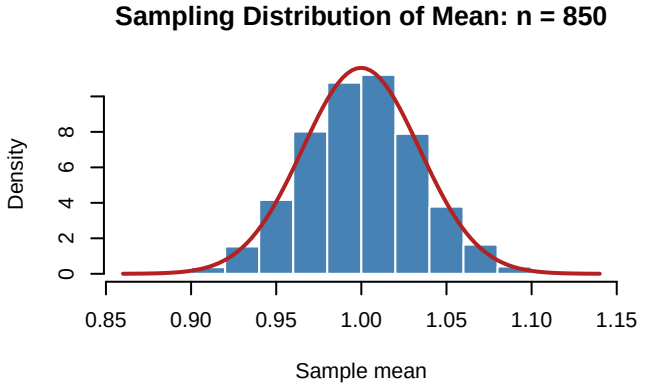

Q−Q Plot: n = 700
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles

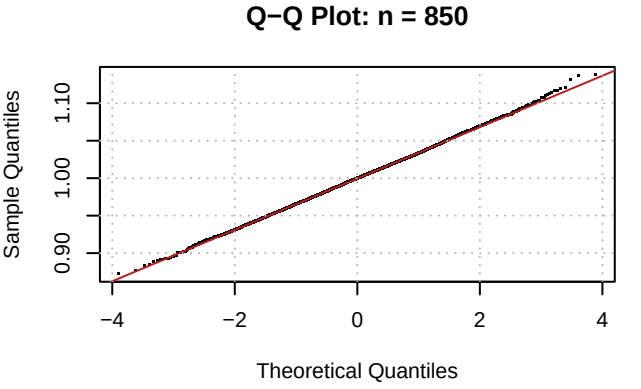

Sampling Distribution of Mean: n = 750
Density
0
2
4
6
8
10
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

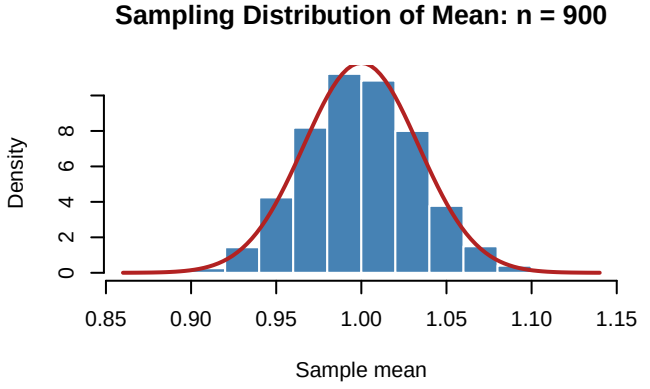

Q−Q Plot: n = 750
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles

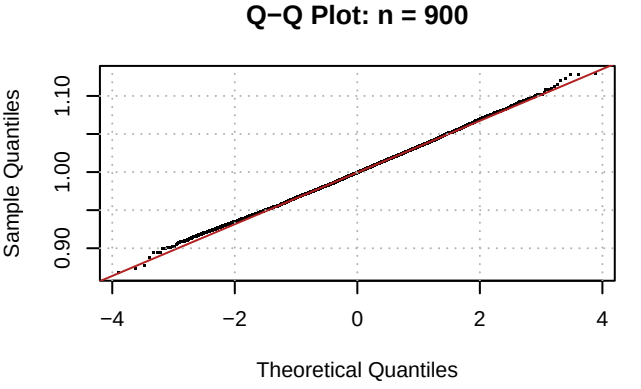

Sampling Distribution of Mean: n = 800
Density
0
2
4
6
8
10
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

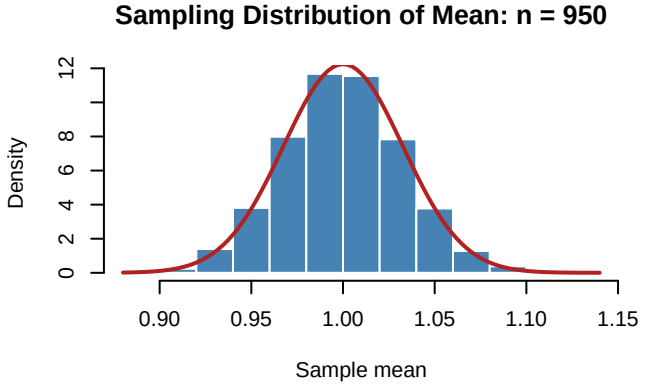

Q−Q Plot: n = 800
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles

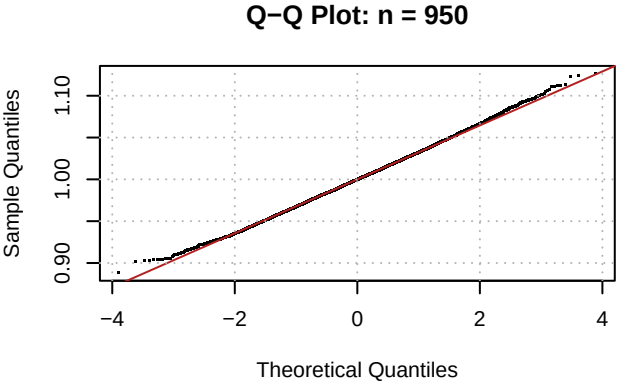

Sampling Distribution of Mean: n = 850
Density
0
2
4
6
8
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

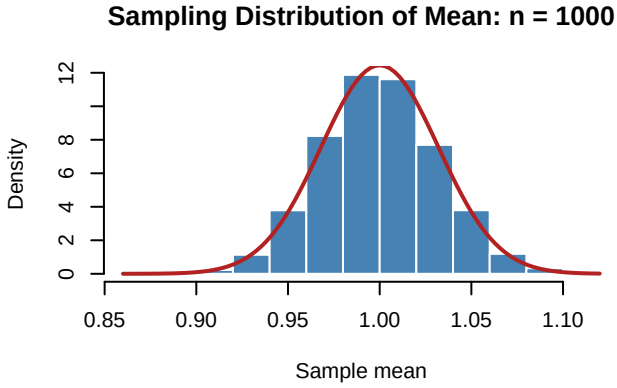

Q−Q Plot: n = 850
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles

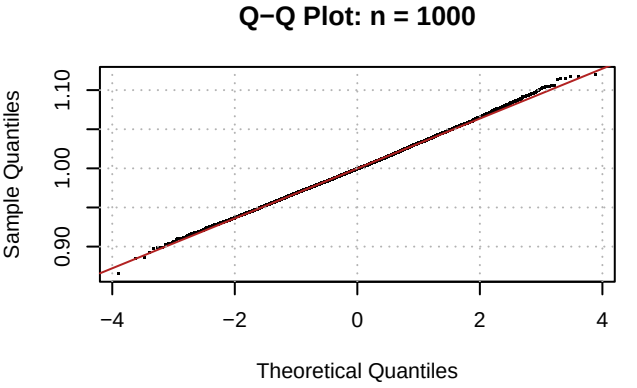

Sampling Distribution of Mean: n = 900
Density
0
2
4
6
8
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

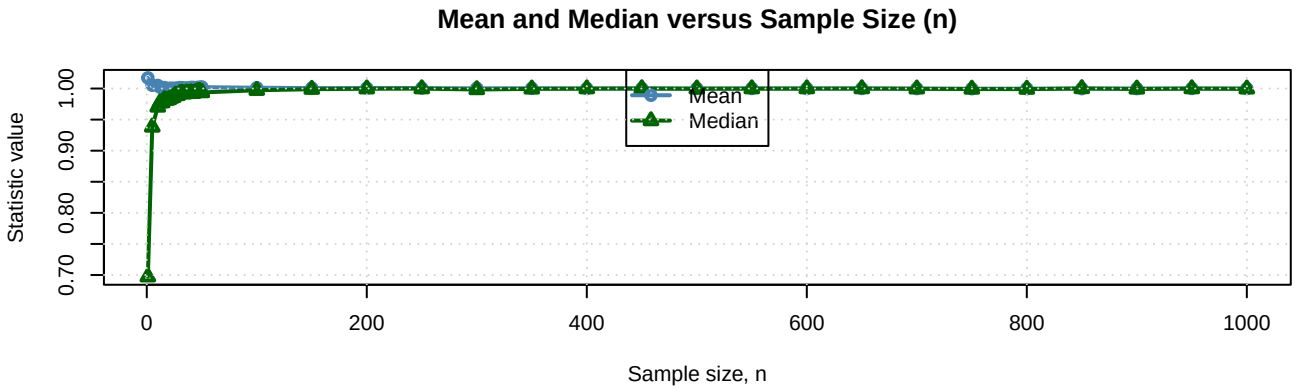

Q−Q Plot: n = 900
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles

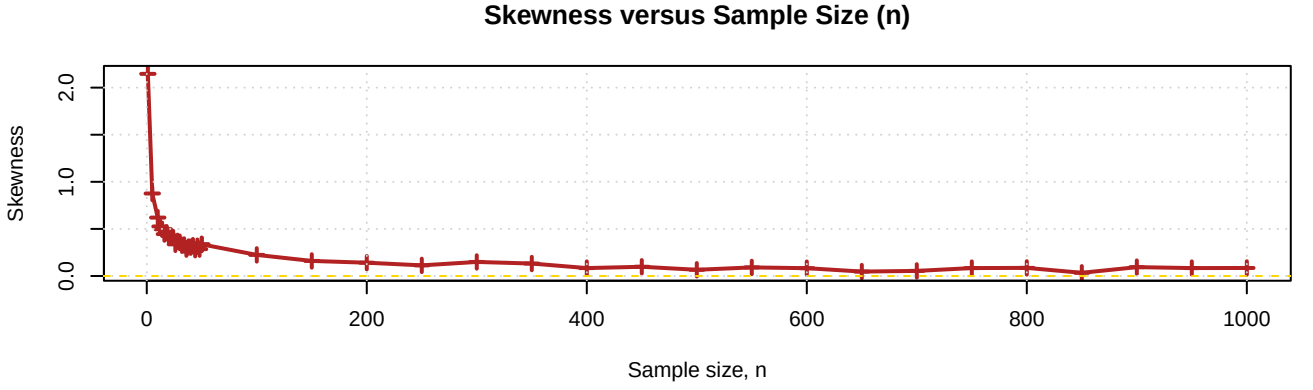

Sampling Distribution of Mean: n = 950
Density
0
2
4
6
8
12
0.90
0.95
1.00
1.05
1.10
1.15
Sample mean

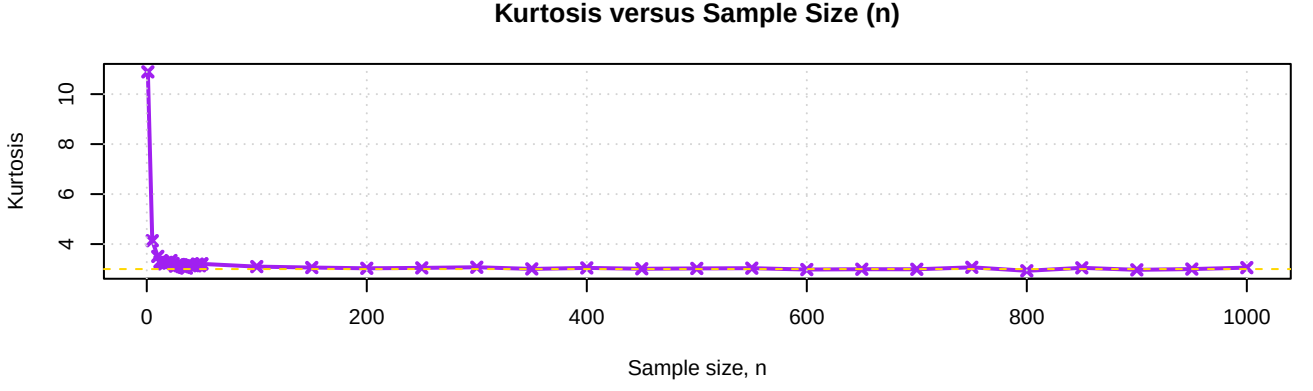

Q−Q Plot: n = 950
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 1000
Density
0
2
4
6
8
12
0.85
0.90
0.95
1.00
1.05
1.10
Sample mean


Q−Q Plot: n = 1000
Sample Quantiles
0.90
1.00
1.10
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Statistic value
0.70
0.80
0.90
1.00
Mean
Median
0
200
400
600
800
1000
Sample size, n


Skewness versus Sample Size (n)
Skewness
0.0
1.0
2.0
0
200
400
600
800
1000
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
4
6
8
10
0
200
400
600
800
1000
Sample size, n